Enumerating separable derangements

Robert Dougherty-Bliss* Alejandro B. Galván* Michaela Polley* David Shuster*

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Abstract

We give a polynomial-time algorithm to compute the number b_n of separable derangements of $[n]$. This algorithm is based on a generating function technique which tracks permutations along with their occupied diagonals, where each permutation is counted once for every such diagonal. We provide bounds for the proportion of separable permutations which are derangements, show that b_n and the large Schröder numbers have the same exponential growth constant $(3 + 2\sqrt{2})$, and use the first 3000 terms to conjecture more explicit asymptotic behavior. This partially answers several questions about separable derangements recently posed by Vatter.

1 Introduction

Enumerating permutations of $[n] = \{1, 2, 3, \ldots, n\}$ which satisfy a given set of restrictions is a fundamental problem in combinatorics. Vince Vatter recently posed the question of finding the generating function for the number of *separable derangements* of $[n]$ [13, Problem 4.4]. This work partially answers this question.

Theorem 1. *There exists a polynomial-time algorithm to compute b_n, the number of separable derangements of $[n]$.*

At the time of writing, Vatter and Jay Pantone computed b_n through $n = 18$ using a method based on "displacement sets" (see [11]):

$$0, 1, 2, 7, 30, 124, 560, 2610, 12470, 60955, 302930, 1528621, 7790780,$$
$$40202007, 208787486, 1094575377, 5766892092, 30609900691, \ldots$$

We can compute through $n = 3000$ using an entirely different approach based on generating functions. For example,

$$b_{100} = 1109494928577851681540444606156833022733492291110692095509038076266474501.$$

These generating functions, which we call "diagonal generating functions," track occupied diagonals in permutations. This leads to, as Wilf put it, an "effective" answer which can stand in "polite society" [15].

Independently, the restrictions "separable" and "derangement" are well-understood. A permutation is a derangement if it has no fixed points. The number of derangements of $[n]$ is

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

*Dartmouth College, Department of Mathematics, Hanover, NH, USA

This can be obtained with inclusion-exclusion or by solving the recurrence $D_n = (n-1)(D_{n-1} + D_{n-2})$. Separable permutations are the smallest class of permutations which contain the identity and are closed under the operations "sum" and "skew sum":

$$(\sigma, \tau) \in S_n \times S_m$$
$$(\sigma \oplus \tau)(i) = \begin{cases} \sigma(i) & i \leq n \\ n + \tau(i-n) & n < i \leq n+m \end{cases}$$
$$(\sigma \ominus \tau)(i) = \begin{cases} \sigma(i) + m & i \leq n \\ \tau(i-n) & n < i \leq n+m. \end{cases}$$

The number s_n of separable permutations is given by the *large Schröder numbers*, which for $n \geq 1$ have the explicit formula

$$s_{n+1} = \frac{1}{n} \sum_{k=1}^{n} 2^k \binom{n}{k} \binom{n}{k-1}.$$

The generating function for s_n is algebraic:

$$\sum_{n=1}^{\infty} s_n x^n = \frac{1 - x - \sqrt{1 - 6x + x^2}}{2}.$$

This generating function can be obtained by applying the method of Brignall, Huczynska, and Vatter [3]. The method constructs generating functions which correspond to permutation families which contain only finitely many "simple" permutations, subject to the restrictions of finitely many "query-complete" sets.[1] It is well-known that the separable permutations are precisely $\mathrm{Av}(2413, 3142)$,[2]and that the only simple separable permutations are 1, 12, and 21 [2, 14], so the method works.

When we combine the restrictions "separable" and "derangement," the situation becomes more delicate.[3]Intuitively, the difficulty is that pattern avoidance is about relative positions while fixed points are about absolute positions. There is no obvious relation between these two that enables one to say when a derangement would be separable or vice versa. The general method from [3] will not help because "being a derangement" is not a finite collection of query-complete sets.

From a more elementary perspective, the separable structure is not that useful when checking if something is a derangement. While the direct sum is simple—$\sigma \oplus \tau$ is a derangement if and only if σ and τ are—the skew sum is not: the main diagonal of $\sigma \ominus \tau$ may intersect σ or τ depending on their sizes. The diagonals of separable permutations must be "tracked" if we are to determine whether a given skew sum is a derangement.

In the following sections we will outline a procedure to calculate the number of separable derangements of $[n]$. The basic idea is to introduce a generating function based on the bivariate sequence

$$a_{nk} = \text{number of separable } \sigma \in S_n \text{ with an occupied } k\text{th diagonal.}$$

This will allow us to count the separable derangements by complementing. In addition, Section 6 contains some results about the asymptotics of b_n compared to s_n.

2 Background: Decomposition of Separable Permutations

Definition 2. A permutation is *sum decomposable* (resp. *skew-sum decomposable*) if it can be expressed as the sum (resp. skew sum) of two nonempty permutations.

[1]The terms "simple" and "query-complete" are defined in [3].

[2]Given a set of patterns $\mathcal{S}$, the set $\mathrm{Av}(\mathcal{S})$ is the set of all permutations which avoid the patterns in $\mathcal{S}$.

[3]The first result about derangements that avoid patterns was by Robertson, Saracino, and Zeilberger, who enumerated derangements in $\mathrm{Av}(\beta)$ for $\beta \in \{132, 213, 321\}$ [12]. Elizalde later extended these results and gave some bijective proofs with Pak [4, 5, 6].

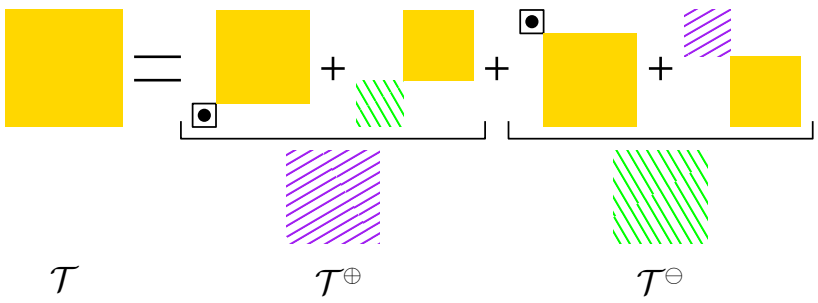


Figure 1: Unique decompositions of sum decomposable and skew-sum decomposable separable permutations.

Definition 3. Define the following objects:

- $\mathcal{T}$ is the set of separable permutations, with ordinary generating function
$$T(x) = \sum_{n=1}^{\infty} s_n x^n.$$
- $\mathcal{T}^{\oplus}$ is the set of sum decomposable separable permutations, with ordinary generating function
$$T^{\oplus}(x) = \sum_{n=1}^{\infty} s_n^{+} x^n.$$
- $\mathcal{T}^{\ominus}$ is the set of skew-sum decomposable separable permutations, with ordinary generating function
$$T^{\ominus}(x) = \sum_{n=1}^{\infty} s_n^{-} x^n.$$

Every sum decomposable separable permutation σ can be uniquely decomposed as $\sigma = \rho \oplus \tau$, subject to the condition that $\tau \in \mathcal{T}$ and either $\rho = 1$ or $\rho \in \mathcal{T}^{\ominus}$. Indeed, this decomposition is obtained by taking the ρ of minimum length such that there exists τ with $\sigma = \rho \oplus \tau$. The minimality condition ensures that $\rho \notin \mathcal{T}^{\oplus}$; and if we picked any larger ρ' such that $\sigma = \rho' \oplus \tau'$, then necessarily ρ' is sum decomposable as $\rho' = \rho \oplus \alpha$ for some α.

Similarly, every skew-sum decomposable, separable permutation σ can be uniquely decomposed as $\sigma = \rho \ominus \tau$, where $\tau \in \mathcal{T}$ and either $\rho = 1$ or $\rho \in \mathcal{T}^{\oplus}$. These decompositions are illustrated in Figure 1.

Theorem 4. *The ordinary generating functions satisfy*

$$\begin{cases} T(x) = x + T^{\oplus}(x) + T^{\ominus}(x), \\ T^{\oplus}(x) = (x + T^{\ominus}(x))T(x), \\ T^{\ominus}(x) = (x + T^{\oplus}(x))T(x). \end{cases} \tag{1}$$

For a proof of this theorem that uses the unique sum and skew-sum decompositions, see [1, Section 2]. Note that their convention includes the empty permutation; replace their $S(x)$ with $1 + T(x)$ to obtain the same system.

3 Diagonal Generating Functions

The diagonals of a permutation refer to the diagonals in its graph as a function. For a permuation of $[n]$, we number the diagonals from $-(n-1)$ to $(n-1)$, starting from the top-left, so that the main diagonal has $k = 0$ (see Figure 2). In other words, $\sigma \in S_n$ has an entry on diagonal k if $i - \sigma(i) = k$ for some $i \in [n]$.

In order to keep track of occupied diagonals of a permutation, we define a new bivariate generating function called the *diagonal generating function*.

Definition 5. Let $\mathcal{C}$ be a set of permutations. Then, the *diagonal generating function* for $\mathcal{C}$ is given by

$$C(x, q) = \sum_{n,k} a_{n,k} x^n q^k$$

where $a_{n,k}$ is the number of permutations in $\mathcal{C}$ of length n that have at least one point on diagonal k.

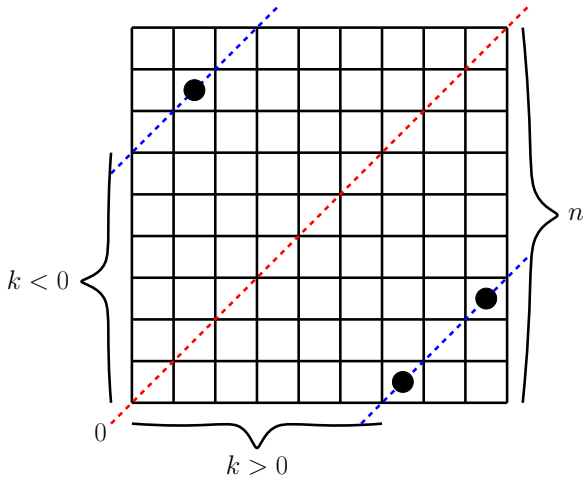


Figure 2: Diagonals as indexed by Definition 5. A permutation of $[n]$ in $\mathcal{C}$ with occupied diagonal k contributes a term of $x^n q^k$ to the diagonal generating function of $\mathcal{C}$.

For example, let $\mathcal{C} = \bigcup_{n \in \mathbb{N}} S_n$. The diagonal generating function starts

$$C(x, q) = q^0 x^1 + \left(q^{-1} + q^0 + q^1\right) x^2 + \left(2q^{-2} + 3q^{-1} + 4q^0 + 3q^1 + 2q^2\right) x^3 + \ldots,$$

as shown in Figure 3.

It is important to note that, in general, the diagonal generating function for $\mathcal{C}$ is not a refinement of the ordinary generating function for $\mathcal{C}$. That is, there is no value that we can substitute for q which always recovers the ordinary generating function. This is because the different permutations in S_n are counted a different number of times. For example, the identity permutation, $12 \ldots n$, is counted exactly once since all of its points lie on the main diagonal; however, the anti-identity permutation, $n \ldots 21$, is counted n times since all of its points lie on different diagonals.

4 Separable Derangements

In this section, we use the structure of separable permutations to deduce a system of equations of generating functions.

Definition 6. Define the following objects:

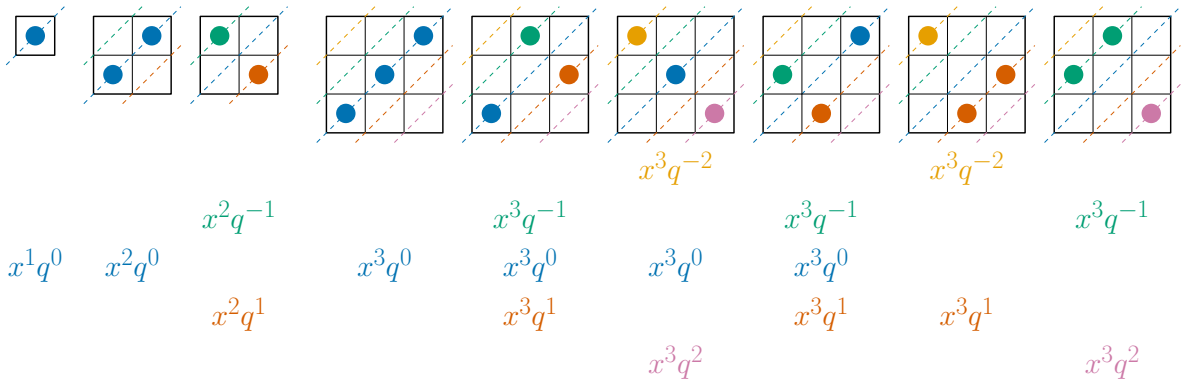


Figure 3: Permutations and the terms they contribute to diagonal generating functions.

- $T(x,q) = \sum_{n,k} a_{n,k} x^n q^k$ is the diagonal generating function for separable permutations.
- $T^{\oplus}(x,q) = \sum_{n,k} a^{+}_{n,k} x^n q^k$ is the diagonal generating function for sum decomposable separable permutations.
- $T^{\ominus}(x,q) = \sum_{n,k} a^{-}_{n,k} x^n q^k$ is the diagonal generating function for skew-sum decomposable separable permutations.

All coefficients are understood to vanish outside of their natural supports. For example, $a_{n,k} = 0$ if $|k| \geq n$ or $n < 0$.

Our objective is to express $T(x,q)$, $T^{\oplus}(x,q)$ and $T^{\ominus}(x,q)$ in terms of each other, $T(x)$, $T^{\oplus}(x)$ and $T^{\ominus}(x)$, obtaining a system of equations that implicitly determines $T(x,q)$. For this purpose, we will use the same structural decomposition as in Section 2, but in this instance we will need to keep track of the diagonals.

Since $\mathcal{T}^{\oplus}$ and $\mathcal{T}^{\ominus}$ are disjoint, and the only element of $\mathcal{T}$ that does not belong to either of them is the permutation of length 1, we have

$$T(x,q) = xq^0 + T^{\oplus}(x,q) + T^{\ominus}(x,q). \tag{2}$$

In order to obtain an equation for $T^{\ominus}(x,q)$, one needs to understand how the skew-sum operation affects the numbering of the diagonals. Let ρ, τ be two permutations of lengths i and k, and let $\sigma = \rho \ominus \tau$. Then, σ has length $i+k$. The j-th diagonal of ρ corresponds to the $(j-k)$-th diagonal of σ, and the ℓ-th diagonal of τ corresponds to the $(\ell+i)$-th diagonal of σ, as shown in Figure 4.

As explained in Section 2, a permutation $\sigma \in \mathcal{T}^{\ominus}$ can be decomposed uniquely as $\rho \ominus \tau$, where ρ is either 1 or an element of $\mathcal{T}^{\oplus}$, and τ is in $\mathcal{T}$. Let i and k be the lengths of ρ and τ. Then, there are two ways to choose a nonempty diagonal of σ:

- Choosing a nonempty diagonal of ρ. Let j be the index of that diagonal in ρ. Then σ with that marked diagonal contributes a term $x^{i+k}q^{j-k}$ to $T^{\ominus}(x,q)$. The total contribution of this case when $\rho = 1$ is

$$\sum_k s_k x^{k+1} q^{0-k} = xq^0 T\left(\frac{x}{q}\right).$$

  And the total contribution of this case when $\rho \in \mathcal{T}^{\oplus}$ is

$$\begin{aligned}\sum_{i,k,j} a^{+}_{i,j} s_k x^{i+k} q^{j-k} &= \sum_{i,j} a^{+}_{i,j} x^i q^j \sum_k s_k x^k q^{-k} \\ &= T^{\oplus}(x,q) T\left(\frac{x}{q}\right).\end{aligned}$$

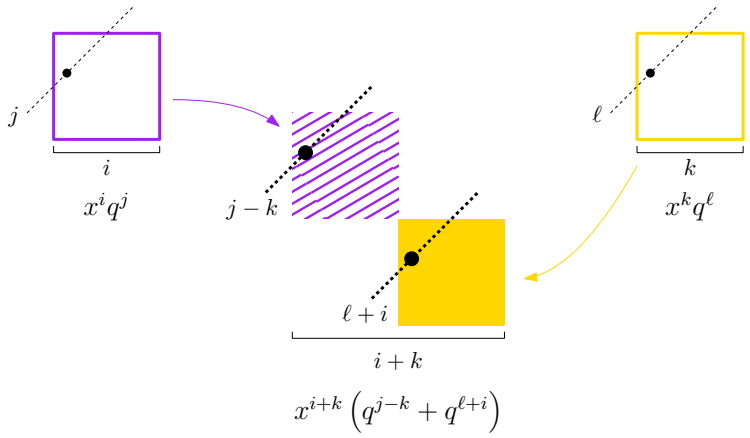


Figure 4: If ρ and τ have lengths i and k, then $\sigma = \rho \ominus \tau$ has length $i+k$. The j-th diagonal of ρ corresponds to the $(j-k)$-th diagonal of σ, and the ℓ-th diagonal of τ corresponds to the $(\ell+i)$-th diagonal of σ.

- Choosing a nonempty diagonal of τ. Let ℓ be its index in τ. Then, σ with that marked diagonal contributes a term $x^{i+k}q^{\ell+i}$ to $T^{\ominus}(x,q)$. The total contribution of this case when $\rho = 1$ is

$$\sum_{k,\ell} a_{k,\ell} x^{k+1} q^{\ell+1} = xqT(x,q).$$

  And the total contribution of this case when $\rho \in \mathcal{T}^{\oplus}$ is

$$\begin{aligned}\sum_{i,k,\ell} s_i^+ a_{k,\ell} x^{i+k} q^{\ell+i} &= \sum_i s_i^+ x^i q^i \sum_{k,\ell} a_{k,\ell} x^k q^\ell \\ &= T^{\oplus}(xq)T(x,q).\end{aligned}$$

These cases are disjoint: diagonals from ρ have indices at most $i-k-1$, and diagonals from τ have indices at least $i-k+1$.

Hence,

$$T^{\ominus}(x,q) = \left(xq^0 + T^{\oplus}(x,q)\right) T\left(\frac{x}{q}\right) + \left(xq + T^{\oplus}(xq)\right) T(x,q). \tag{3}$$

In the case of $T^{\oplus}(x,q)$, the sum operation does not affect the numbering of the diagonals. Indeed, let ρ, τ be two permutations of lengths i and k, and let $\sigma = \rho \oplus \tau$. Then, σ has length $i+k$, the j-th diagonal of ρ corresponds to the j-th diagonal of σ, and the ℓ-th diagonal of τ corresponds to the ℓ-th diagonal of σ, as shown in Figure 5.

As explained in Section 2, a permutation $\sigma \in \mathcal{T}^{\oplus}$ can be decomposed uniquely as $\rho \oplus \tau$, where ρ is either 1 or an element of $\mathcal{T}^{\ominus}$, and τ is in $\mathcal{T}$. Let i and k be the lengths of ρ and τ. Then, there are two ways to choose a nonempty diagonal of σ:

- Choosing a nonempty diagonal of ρ, with index j. This contributes a term $x^{i+k}q^j$ to $T^{\oplus}(x,q)$. The total contribution of this case is

$$\sum_k s_k x^{k+1} q^0 + \sum_{i,k,j} a^-_{i,j} s_k x^{i+k} q^j = \left(xq^0 + T^{\ominus}(x,q)\right) T(x).$$

- Choosing a nonempty diagonal of τ, with index ℓ. This contributes a term $x^{i+k}q^{\ell}$ to $T^{\oplus}(x,q)$. The total contribution of this case is

$$\sum_{k,\ell} a_{k,\ell}x^{k+1}q^{\ell} + \sum_{i,k,\ell} s_i^- a_{k,\ell}x^{i+k}q^{\ell} = \left(x + T^{\ominus}(x)\right) T(x,q).$$

However, these two cases are not disjoint. To account for the double counting, one needs to subtract the cases corresponding to diagonals that are nonempty in both ρ and τ. This correction term is given by

$$\sum_{k} a_{k,0}x^{k+1}q^{0} + \sum_{i,k,j} a_{i,j}^- a_{k,j}x^{i+k}q^{j} = \left(xq^0 + T^{\ominus}(x,q)\right) \bigodot_q T(x,q),$$

where $\bigodot_q$ denotes the Hadamard product (termwise multiplication) with respect to q. By that, we mean

$$\left(\sum_j f_j(x)q^j\right) \bigodot_q \left(\sum_j g_j(x)q^j\right) = \sum_j f_j(x)g_j(x)q^j.$$

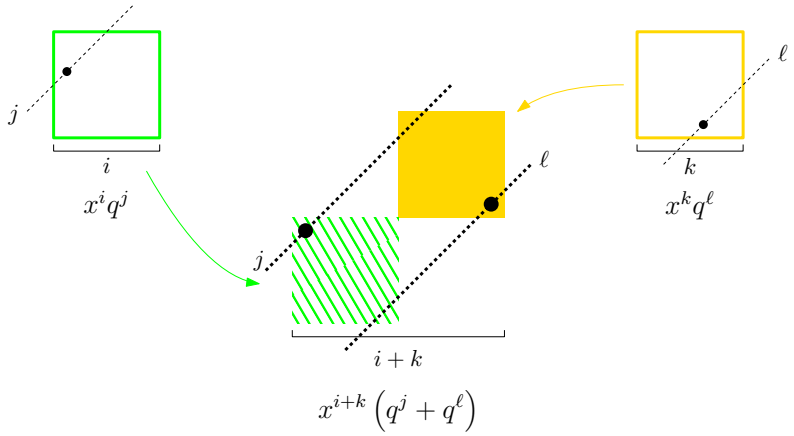


Figure 5: If ρ and τ have lengths i and k, then $\sigma = \rho \oplus \tau$ has length $i+k$. The j-th diagonal of ρ and ℓ-th diagonal of τ correspond to the j-th and ℓ-th diagonals of σ.

Hence,

$$T^{\oplus}(x,q) = \left(xq^0 + T^{\ominus}(x,q)\right) T(x) + \left(x + T^{\ominus}(x)\right) T(x,q) - \left(xq^0 + T^{\ominus}(x,q)\right) \bigodot_q T(x,q). \tag{4}$$

Theorem 7. *The functions $T(x,q)$, $T^{\oplus}(x,q)$ and $T^{\ominus}(x,q)$ satisfy the following system:*

$$\begin{cases} T(x,q) & = xq^0 + T^{\oplus}(x,q) + T^{\ominus}(x,q), \\ T^{\ominus}(x,q) & = \left(xq^0 + T^{\oplus}(x,q)\right) T\left(\frac{x}{q}\right) + \left(xq + T^{\oplus}(xq)\right) T(x,q), \\ T^{\oplus}(x,q) & = \left(xq^0 + T^{\ominus}(x,q)\right) T(x) + \left(x + T^{\ominus}(x)\right) T(x,q) - \left(xq^0 + T^{\ominus}(x,q)\right) \bigodot_q T(x,q). \end{cases} \tag{5}$$

5 Recurrences and implementation

In this section we will give explicit recurrences implied by (5), then explain some implementation details about how terms were computed. Refer to Definition 6 for the meaning of each sequence. The following formulas use *Iverson bracket* notation from [8], defined as

$$[P] = \begin{cases} 1 & \text{if } P \text{ is true} \\ 0 & \text{if } P \text{ is false} \end{cases}$$

for any statement P.

Recall that our goal is to compute $s_n - a_{n,0}$, the number of separable derangements. Extracting coefficients on x^n in (1) and $x^n q^k$ in (5), followed by some rearranging, gives a set of interlocking recurrences. We will give them in two parts.

Proposition 8. *The sequences s_n, s_n^+, and s_n^- satisfy the recurrences*

$$\begin{aligned} s_n &= s_n^+ + s_n^- \\ s_n^+ &= \sum_{i=1}^{n-1} ([i=1] + s_i^-) s_{n-i} \\ s_n^- &= \sum_{i=1}^{n-1} ([i=1] + s_i^+) s_{n-i} \end{aligned}$$

for $n \geq 2$ with initial conditions $(s_1, s_1^+, s_1^-) = (1, 0, 0)$.

Proposition 9. *The sequences $a_{n,k}$, $a_{n,k}^+$, and $a_{n,k}^-$ satisfy the recurrences*

$$\begin{aligned} a_{n,k} &= a_{n,k}^+ + a_{n,k}^- \\ a_{n,k}^+ &= \sum_{i=1}^{n-1} \Big(([i=1][k=0] + a_{i,k}^-)(s_{n-i} - a_{n-i,k}) + ([i=1] + s_i^-) a_{n-i,k} \Big) \\ a_{n,k}^- &= \sum_{i=1}^{n-1} \Big(([i=1][k+n-i=0] + a_{i,k+n-i}^+) s_{n-i} + ([i=1] + s_i^+) a_{n-i,k-i} \Big) \end{aligned} \tag{6}$$

for $n \geq 2$ with initial conditions $(a_{1,k}, a_{1,k}^+, a_{1,k}^-) = ([k=0], 0, 0)$.

5.1 Implementation details and timing

It is straightforward to implement these recurrences on a computer. We first implemented them in Maple by hand, then translated them to C with GPT-5.6 Sol. The FLINT library was used to handle large-integer arithmetic [7].

Below is a rough upper-bound estimate on the runtime of computing with the recurrences (6). It accounts for the cost of large-integer arithmetic.

Theorem 10. *The numbers $a_{i,k}$ with $1 \leq i \leq n$ and $|k| < i$ can be computed with $O(n^3 M(n))$ bit operations, where $M(n)$ is the number of bit operations needed to multiply integers with n bits.*

Proof. The recurrences (6) compute $a_{n,k}$ by "row." Once the previous rows are known, (6) requires $O(n)$ additions and multiplications. There are $O(n)$ entries, so the nth row takes $O(n^2 M(\alpha_n))$, where α_n is an upper bound for the bit-lengths of the $a_{i,k}$ for $i < n$.

The asymptotics of the Schröder numbers s_n are well-known [10] and imply that the bit-length of s_n is $O(n)$. Since $0 \leq a_{n,k}, a_{n,k}^{\pm} \leq s_n$, the same is true for $a_{n,k}$ and $a_{n,k}^{\pm}$, so we can take $\alpha_n = O(n)$. Therefore the

n	Median time (s)	Time range (s)
100	0.01	0.01–0.04
200	0.03	0.03–0.03
300	0.08	0.07–0.08
400	0.25	0.22–0.25
500	0.70	0.69–0.72
600	1.53	1.51–1.70
700	2.85	2.85–2.89
800	4.91	4.85–4.95
900	7.92	7.92–7.99
1000	12.33	12.21–12.89

Table 1: Time to compute $a_{n,0}$ for different n. Each row reports the median and range of three fresh runs. The benchmark was performed on an AMD Ryzen AI 7 445, a 6-core CPU with 12 threads.

nth row costs $O(n^2 M(n))$. Summing over all rows, and assuming that $M(n)$ is increasing, gives the total cost

$$\sum_{i=1}^{n} O(i^2 M(i)) \leq M(n) \sum_{i=1}^{n} O(i^2) = O(n^3 M(n)). \qquad \square$$

There are a number of performance improvements to be made for the recurrences. First, by applying the inversion bijection we see that $a_{n,k} = a_{n,-k}$ for all k, so it is unnecessary to compute $a_{n,k}$ for $k < 0$. The same is true for $a^+_{n,k}$ and $a^-_{n,k}$. Second, the computation of $a_{n,k}$ depends only on values of $a_{i,k}$ with $i < n$; in other words, every entry in a new row can be computed independently in parallel. Finally, if the goal is to only compute $a_{n,k}$ for some large n, then it is unnecessary to compute all $a_{i,j}$ with $1 \leq i \leq n$ and $0 \leq j < i$. The recurrence (6) shows that $a_{n,k}$ only depends immediately on three lines of terms:

1. $a_{i,k}$ for $i < n$
2. $a_{i,k+n-i}$ for $(k+n)/2 < i < n$
3. $a_{n-i,k-i}$ for $i < (k+n)/2$

Repeating this idea on the points on these lines shows that only a "diamond" of terms need to be computed to determine $a_{n,0}$. (Or a half-diamond if the symmetry optimization $a_{n,k} = a_{n,-k}$ is used.) These performance optimizations give constant-factor improvements. The asymptotic runtime is unchanged from $O(n^3 M(n))$. Experimental runtime data from our C implementation is in Table 1.

There is one more optimization, though it is slightly more involved. Rather than tracking *occupied* diagonals, which leads to (6), it is computationally simpler to track *empty* diagonals. Diagonal k is empty in $\sigma \oplus \tau$ if and only if it is empty in σ and τ. This eliminates the need for inclusion-exclusion in (6), which reduces the number of multiplications in the equation for $a^+_{n,k}$. A similar optimization works for $a^-_{n,k}$. This conceptual change does not change the asymptotic runtime, nor does it lead to any significantly different generating function equations.

6 Asymptotics

In [13, Question 4.5], Vatter asks the following question: What is the limiting proportion of derangements among separable permutations, assuming this limit exists? In this section we will provide some partial answers.

Vatter originally suggested that b_n/s_n looks to be monotonically decreasing, and perhaps converging to a constant in $[1/5, 1/4]$. We observe that b_n/s_n is monotonically decreasing for $5 \leq n \leq 3000$, but it seems to decrease to 0. See Figure 6.

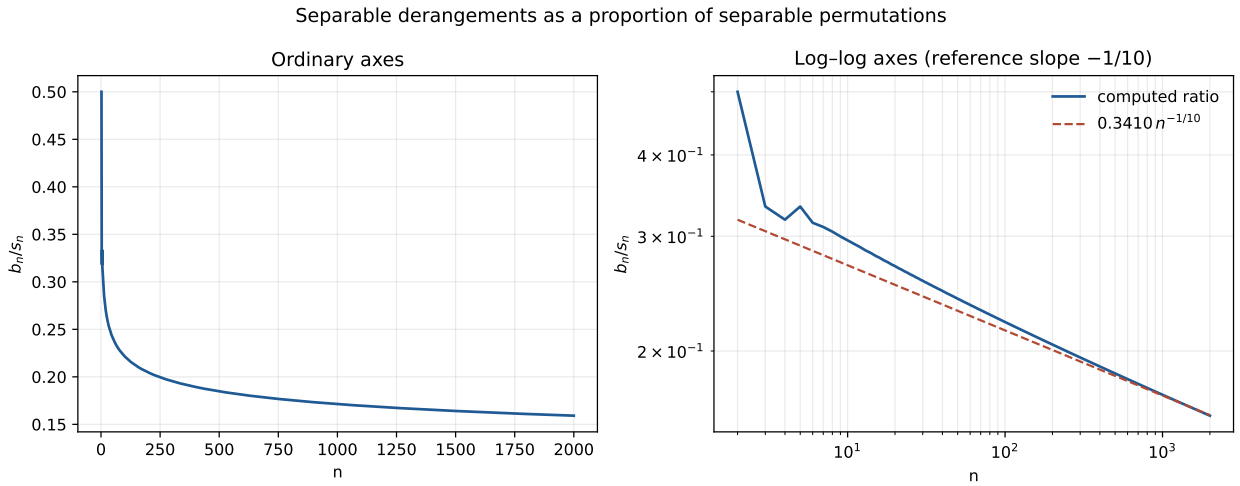


Figure 6: The ratio b_n/s_n up to $n = 2000$. On the right, the dotted line shows the projected asymptotics if Conjecture 14 were true with $\alpha = 1.6$. The intercept was fit with numerical data.

The following results put some theoretical limits on how rare derangements can be among the separable permutations.

Lemma 11. *For any integer $n \geq 1$ we have $b_{2n} \geq s_n^2$ and $b_{2n+1} \geq s_n(s_{n+1} - s_n)$. For $n \geq 3$ we have $b_n \leq s_n - 2s_{n-1} + s_{n-2}$.*

Proof. If $\sigma, \tau \in S_n$ are any permutations of length n, then $\sigma \ominus \tau$ is a derangement. In particular, this is also true if σ and τ are separable, so there are at least s_n^2 separable derangements of length $2n$.

Let $\sigma \in S_n$ and $\tau \in S_{n+1}$ be separable. Then $\sigma \ominus \tau$ is a derangement if and only if $\tau(1) \leq n$, meaning that τ does not begin with its maximum. Any separable permutation which begins with its maximal value is $1 \ominus \tau'$ for some separable τ' of length one less, and so there are exactly s_n different τ which would not create a derangement. Therefore, there are at least $s_n(s_{n+1} - s_n)$ derangements of length $2n + 1$.

For the upper bound, note that no separable derangement of $[n]$ starts with 1 or ends with n. There are s_n separable permutations; s_{n-1} of them start with 1, s_{n-1} of them end in n, and s_{n-2} of them do both. By inclusion-exclusion, $b_n \leq s_n - 2s_{n-1} + s_{n-2}$. □

For our later results, it is convenient to recall the asymptotics of the Schröder numbers [10]:

$$s_n \sim C\lambda^n n^{-3/2} \tag{7}$$

where C is a positive constant and

$$\lambda = 3 + 2\sqrt{2}.$$

Theorem 12. *There exists a constant $\beta > 0$ such that*

$$\beta n^{-3/2} \leq \frac{b_n}{s_n} \leq 0.69$$

for sufficiently large n.

Proof. By Lemma 11, we have

$$\frac{b_{2n}}{s_{2n}} \geq \frac{s_n^2}{s_{2n}} \sim C\left(\frac{2}{n}\right)^{3/2}.$$

This implies $b_{2n}/s_{2n} = \Omega(n^{-3/2})$. Similarly,

$$\frac{b_{2n+1}}{s_{2n+1}} \geq \frac{s_n(s_{n+1} - s_n)}{s_{2n+1}} \sim C\left(\frac{2}{n}\right)^{3/2}(1 - \lambda^{-1}).$$

This implies $b_{2n+1}/s_{2n+1} = \Omega(n^{-3/2})$. Taken together, these statements imply $b_n/s_n = \Omega(n^{-3/2})$.

For the other direction, the upper bound part of Lemma 11 implies

$$\frac{b_n}{s_n} \leq 1 - 2\frac{s_{n-1}}{s_n} + \frac{s_{n-2}}{s_n}.$$

Letting $n \to \infty$ in the right-hand side produces

$$1 - \frac{2}{\lambda} + \frac{1}{\lambda^2} = 0.68629\ldots.$$

In particular, $b_n/s_n \leq 0.69$ for sufficiently large n. □

Theorem 13. *Let $\lambda = 3 + 2\sqrt{2}$. Then*

$$\lim_{n\to\infty} b_n^{1/n} = \lim_{n\to\infty} s_n^{1/n} = \lambda.$$

Proof. The Schröder limit is a consequence of the asymptotics (7) of s_n. For b_n, note that we have the inequality

$$s_n^2 \leq b_{2n} \leq s_{2n}.$$

Taking $2n$th roots on both sides shows that $\lim_{n\to\infty} b_{2n}^{1/(2n)} = \lambda$. The odd n case is handled similarly. □

The previous result shows that $b_n > (\lambda - \epsilon)^n$ is eventually true for any $\epsilon \in (0, \lambda)$. In particular, if b_n/s_n goes to 0, it does not do so for "trivial" exponential growth reasons. We conjecture the following.

Conjecture 14. *There exist constants $C' > 0$ and $\alpha \in (3/2, 3]$ such that*

$$b_n \sim C'\frac{\lambda^n}{n^\alpha}.$$

Accordingly,

$$\frac{b_n}{s_n} = \Theta(n^{3/2-\alpha})$$

How would we estimate α if Conjecture 14 were true? To remove C' from the picture, form quotients of terms a fixed-proportion away:

$$\frac{b_{2n}}{b_n} \sim \lambda^n 2^{-\alpha}$$

This relation is equivalent to

$$\log\frac{b_{2n}}{b_n} = n\log\lambda - \alpha\log 2 + o(1),$$

and so

$$\alpha = n\log_2\lambda - \log_2\frac{b_{2n}}{b_n} + o(1). \tag{8}$$

Empirically it seems that (8) is converging to a constant. With $n = 1500$ we estimate $\alpha \approx 1.605665$, and with $n = 1000$ we estimate $\alpha \approx 1.606874$. It could be that $\alpha = 1.6$, which would give $b_n/s_n = \Theta(n^{-1/10})$. Of course, the conjecture could also be false. It would be difficult to empirically distinguish $b_n/s_n = \Theta(n^{-1/10})$ from $b_n/s_n = \Theta((\log n)^{-1})$ without much larger n.

7 Conclusion and Open Questions

Using diagonal generating functions, we have given an explicit polynomial-time algorithm to compute b_n, the number of separable derangements of $[n]$. This allows us to compute significantly more terms than were previously known. We believe that diagonal generating functions may be useful in other contexts. In particular, separable derangements are merely derangements in $\mathrm{Av}(2413, 3142)$; the same ideas may apply to derangements which avoid other patterns. We intend to investigate this in a later publication.

There are a number of remaining questions. First, while we have a polynomial-time algorithm to compute b_n, we have been unable to conjecture an explicit formula, generating function, D-finite (P-recursive) recurrence, or differential equation [9]. There may be a more explicit representation of b_n, but we have not found one.

Second, the true asymptotics of b_n/s_n are still open. Based on our empirical data, it seems certain that $b_n/s_n \to 0$, but we cannot rule out other possibilities.

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